

Faster / simpler $\rightarrow$

**Klein–Gordon /
Helmholtz Equation**

$$[\nabla^2 + k_0^2 n^2(\mathbf{r})] \psi(\mathbf{r}) = 0$$

*Parent relativistic
scalar wave equation*

**Forward KG ODE
Ground Truth**

$$\frac{d}{dz} \begin{pmatrix} \psi \\ \phi \end{pmatrix} = \begin{pmatrix} \phi \\ -(\nabla_\perp^2 + k_0^2 n^2) \psi \end{pmatrix}$$

*Adaptive second-order solve;
no split-step approximation*

WPM (Wide-angle, local n)

$$\begin{aligned} \psi(z + \Delta z) &\approx \sum_m w_m(x, y) \psi_m \\ \psi_m &= \mathcal{F}^{-1} \left[e^{i\Delta z \sqrt{k_0^2 n_m^2 - k_\perp^2}} \mathcal{F}\{\psi(z)\} \right] \end{aligned}$$

*Binned local-index approximation
to the one-way square-root operator*

Angular Spectrum Multislice

$$\psi(z + \Delta z) \approx e^{i\sigma V_{\text{proj}}} \mathcal{F}^{-1} \left[e^{i\Delta z \sqrt{k_0^2 - k_\perp^2}} \mathcal{F}\{\psi(z)\} \right]$$

*Exact free-space dispersion with
standard transmission step*

Fresnel Multislice

$$\psi(z + \Delta z) \approx e^{i\sigma V_{\text{proj}}} \mathcal{F}^{-1} \left[e^{-ik_\perp^2 \Delta z / (2k_0)} \mathcal{F}\{\psi(z)\} \right]$$

*Paraxial limit of the
angular-spectrum kernel*

$\longrightarrow$
choose z as propagation axis
 $\nabla^2 = \partial_z^2 + \nabla_\perp^2$

$\longrightarrow$
one-way branch plus Schmidt
WPM approximation
 $\sqrt{\nabla_\perp^2 + k_0^2 n^2} \rightarrow \sqrt{k_0^2 n_m^2 - k_\perp^2}$

$\longrightarrow$
homogeneous/free-space limit
 $n_m \rightarrow 1$ in the propagation kernel

$\longrightarrow$
small-angle Taylor expansion
 $\sqrt{k_0^2 - k_\perp^2} \approx k_0 - k_\perp^2 / (2k_0)$

$\leftarrow$ More accurate / more general